

ZAYED BIN AHMED

7/9 North Dhanmondi, Dhaka, Bangladesh

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Education

Bangladesh University of Engineering and Technology

March 2018 – April 2023

Bachelor of Science in Mechanical Engineering

Dhaka, Bangladesh

CGPA:3.45/4.00

Thesis: 'Magneto-thermo-elastic Analysis of a Non-uniform Sandwich FGM Rotating Disk Under Thermal Shock'

Supervisor: S Reaz Ahmed

Research Interest

- Computational Physics
- Machine Learning
- FDM
- Composites
- Modelling/Simulation
- Additive Manufacturing
- Structural Physics
- Meta-materials

Research Experience

Transient Magneto-thermoelastic Analysis of FGM Rotating Disk

PI:Dr.Shaikh Reaz Ahmed

- Multiphysics analysis of FGM rotating disk in magnetic field consisting thermoelastic effect on boundary points.
- Sandwich FGM with metal, FGM, ceramic layout under centrifugal force and forced convection.
- Effect of temperature shock on boundary as a function of temperature and its effect on stress.
- To develop a numerical scheme to evaluate the effect of mechanical shock using FDM on MATLAB.

Optimum Drug Level of Chemotherapy using Sliding Mode and Synergetic Controller

PI:Dr.Sumon Saha

- Design the cancer model and different controllers of Drug intake in MATLAB Simulink and simulate for Optimum Drug level for minimum growth of cancer cells.
- Robust Control of Non-linear Brain Cancer Model using Higher Order Sliding Mode and Synergetic Controllers
- Using Simulink to determine the parameters.

Publications

- **Zayed Bin Ahmed**, S Reaz Ahmed, Pranta Rahman Sarkar, '*Magneto-thermo-elastic Analysis of a Non-uniform Sandwich FGM Rotating Disk Under Thermal Shock*', Expected submission in *The Journal of Strain Analysis for Engineering Design* by January 2024. [In preparation]

Projects

Waste Sorting Smart Dustbin | Solidworks, TinkerCAD, Arduino

Jan.2021 - May.2021

- Sensor based smart dustbin to sort different types of wastes i.e. Plastic, Metal, Glass, Paper

Design And Modelling of Plate Heat Exchanger |

Nov.2021-Feb.2022

- Homemade heat exchanger project
- Design a Die-press system in Solidworks; Manufacturing of Die-press using CNC Machine
- Assembled the plates and performed experiments

Technical Skills

- **Programming Languages:** Python, C, HTML, MATLAB
- **Simulation Tools:** MATLAB Simulink, COMSOL, ANSYS
- **CAD softwares:** AutoCAD, SolidWorks
- **Documentation Software:** Microsoft Office Suite, LaTeX, Notepad
- **Graphing Software:** Microsoft Office Suite, Tecplot
- **Other Softwares:** Tinkercad, Engineering Equation Solver (EES)

Professional Experience

GPH Ispat ltd.

November 2022

Engineering Intern

- Learned about Mechanical Maintenance, Product Quality Control, Fire-safety features, management of state-of-the-art Steel building factory.
- Gained firm knowledge of the operation of a Quantum Arc Furnace, Cooling Towers, Air Purification System etc.

Affiliations

BUET Entrepreneurship Development Club

April 2018 - May 2023

Member

BUET

IMechE BUET Student Chapter

May 2019 - May 2023

Affiliate Member

BUET

BUET Badhon(Blood Donation Group)

April 2018 - May 2023

Member

BUET

BUET Titumir Hall)

April 2022 - May 2022

Hall Mess Manager(with over 400 residents)

BUET

Test Scores

GRE: 311 (Quant: 164, Verbal: 147, Analytical Writing: 3.5)

TOEFL:106 (Reading 28; Listening 29; Speaking 25; Writing 24)

Awards and Degrees

Certified SolidWorks Associate

April 2020

Mechanical Design

HSC Scholarship Dhaka Board (General Category)

Nov. 2017

Recognition for excellent result in HSC Examination

SSC Scholarship Chittagong Board (Talent Pool Category)

Aug. 2015

Recognition for excellent result in SSC Examination

References

Dr. Shaikh Reaz Ahmed

Professor, Department of
Mechanical Engineering
BUET,Dhaka

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Dr.Md Jane Alam Khan

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Dr. Sheikh Mohammad Shavik

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